

Banach-Space Norm Descriptors for Interferometric Profile Metrology: Complementing Amplitude-Averaging Parameters

Abstract

Amplitude-averaging surface parameters such as R_a show negligible response to localised defects — a controlled sensitivity experiment demonstrates that a single 100 nm synthetic spike injected into a spherical-artefact profile changes R_a by 0.0%, and the ISO 4287 shape parameters R_{sk} and R_{ku} are similarly insensitive to such symmetric localised features. This paper shows that Banach-space norm descriptors — L^p , Sobolev $W^{1,2}$, and bounded-variation (BV) norms, applied within a metrological processing pipeline — show substantially increased sensitivity to such defects (BV density: +18%; $W^{1,2}$ gradient seminorm: +622%) when computed from the same interferometrically reconstructed profile without additional hardware. A complete pipeline from raw Michelson-interferometer fringe images to nanometre profile descriptors is presented and validated on two JENOPTIK standards (roundness standard FN 111, $\varnothing = 29.9588$ mm, 24,000 frames; magnification standard FN 101 with Flick flat, 100,000 frames), each measured in five 400° sweeps. External validation against 104 tactile reference measurements (Hommel Roundscan 535) yields $E_n = 0.61$ for the FN 111 and 0.3% agreement for the FN 101 (optical–tactile polar comparison: Supplementary Fig. S16), suggesting metrological consistency of the optical pipeline within the stated within-run type-A uncertainties. The norm-based descriptors are presented as complementary to the ISO 4287 baseline: R_{sk} responds to asymmetry, BV density responds to localised gradients, and their combination may provide descriptive sensitivity beyond that of any single classical parameter in the tested scenarios. The findings suggest that supplementing the standard ISO 4287 report with BV density — at negligible computational

cost — may provide a gradient-sensitive complement to amplitude averaging; the L^4/L^1 peakiness ratio offers a supplementary distribution-shape axis with a direct physical interpretation.

Keywords: interferometry, roundness, surface profile, Banach-space norms, bounded variation, Sobolev seminorm, uncertainty

1. Introduction

Surface topography is one of the most important properties of technical parts. The accuracy of form and roughness influences contact conditions, wear and lifetime. Roundness, understood as deviation from a perfect circle, remains a central requirement in precision manufacturing; Zhu et al. emphasised that precise measurements are needed both for form tolerances and for tribological surface texture assessment [1]. At nanometric scale, such measurements require high-precision optical systems [2]. The foundational reference for surface metrology terminology and parameter definitions is the handbook of Whitehouse [3].

Optical dimensional metrology has a long connection with interference [4]. Modern optical surface metrology relies on diffraction, polarisation and interference [5, 6]. Diffraction and scattering have been applied to roughness measurement [7–11], while interference remains central for ultra-precise surface-form measurement [12, 13] and phase-shifting interferometry is widely used in topography evaluation [14–16].

Several interferometric instruments have been proposed for high-accuracy surface and roundness measurement. Polack et al. described a Michelson-based system for surface-shape determination built around a high-resolution stitching interferometer developed by Thomasset et al. [17, 18]. Fan et al. presented a roundness measuring system for microspheres based on two small Michelson interferometers [19]. Kuhnel et al. proposed an interferometric system for roundness and cylindricity measurement of ring gauges with 0.1 nm resolution over a large measurement range [20]. Commercial instruments can be highly accurate, but they are often costly and may not expose the full acqui-

sition chain needed for experimental method development. The present work therefore uses a self-built, non-contact Michelson-type roundness measurement system assembled from off-the-shelf components. The optical setup, apparatus description and radial image-processing route for phase extraction were published in a previous study [21], where only roughness parameters were reported for the ceramic standard FN 111. The present work extends that study in three directions: (i) roundness ($RONt$, $RONq$) is evaluated for the first time with this self-built optical setup, (ii) a second artefact — the JENOPTIK magnification standard FN 101 with a Flick flat — is measured and validated against its DAkkS-DKD certificate, and (iii) the analysis is extended from classical ISO parameters to incorporate gradient-structure norms — L^p , Sobolev $W^{1,2}$, and bounded-variation (BV) — which are standard tools of functional analysis, here applied within an interferometric profile metrology pipeline.

The computational problem is that modern interferometric surface metrology records long image sequences before a numerical profile exists. In the present data, however, each PNG file is already a separate interferogram in which the phase is encoded spatially by the bright and dark fringes inside the image. The files in the FN 111 and FN 101 folders are therefore not treated as a temporal phase-shifting stack of one fixed scene. Instead, each image $I_i(x, y)$ is processed independently to obtain one wrapped excess-fraction value ϵ_i . The ordered sequence of these values is then unwrapped over image order and converted to a nanometre profile. The full measurement archive contains 24,000 grayscale interferograms of the FN 111, with image size 648×488 px, and 100,000 grayscale interferograms of the JENOPTIK magnification standard FN 101, with image size 640×480 px.

The literature relevant to this comparison is mature but only partly connected. Interferometric testing of spherical, flat and non-standard optical surfaces has developed sophisticated approaches for stitching, registration, motion-error compensation, phase-map reconstruction and systematic-error control [22–26]. Repeatability in interferometric measurement is commonly evaluated through RMS/PV quantities, repeated height-map statistics or pixel-wise standard-deviation

matrices [27, 28]. In uncertainty-oriented surface metrology, surface-texture parameters can be embedded in GUM-style propagation through metrological characteristics and topography-fidelity contributions [29]. Functional data analysis provides a statistical language for curves, images and higher-dimensional functions, including mean functions, amplitude variation, phase variation and functional principal components [30, 31]. Surface topology and scale-dependent roughness studies show that characterisation can depend on persistent spatial features, bandwidth and filtering scale rather than only on field parameters [32–34].

The research gap is therefore specific. Existing interferometric literature provides robust instrumentation, radial fringe processing, phase reconstruction and best-fit-reference handling; standards-oriented work provides scalar profile, form and uncertainty language; and functional data analysis provides a mathematical framework for function-valued observations. What remains insufficiently tested is a complete workflow that begins with raw interferograms, keeps the per-image phase decoding explicit, converts the unwrapped excess-fraction sequence to height using the reflection half-wavelength convention, removes object-specific form and eccentricity, and then computes classical roughness/roundness quantities and gradient-based functional descriptors from the same nanometre residual profile. The classical side of the comparison is restricted to profile and form quantities such as R_a , R_q , R_z , $RONt$ and $RONq$. Areal parameters are not reported because the single-image radial extraction does not produce a calibrated areal surface-topography measurement. The aim is to assess whether gradient-structure descriptors — the L^p norm family, bounded-variation density, and Sobolev seminorms — offer complementary descriptive sensitivity alongside ISO 4287 amplitude and shape parameters when applied to the same reconstructed residual profile, and to evaluate the within-run repeatability of all descriptors across the five recorded sweeps.

84 2. Experimental Details

85 2.1. Optical Setup

86 The optical layout of the measurement system is shown in Fig. 1. The
87 system is a two-channel modified Michelson interferometer with channels shifted
88 by approximately 90° in phase. When an array detector is used instead of a
89 point detector, the second channel is not strictly required [35]. In this system it
90 was retained because it is useful for object centring and can serve as a control
91 channel.

92 The optical setup and the full apparatus-level description were reported in a
93 previous study on radial image processing for phase extraction in rough-surface
94 interferometry [21]. They are repeated here in condensed form to make the
95 present functional-descriptor analysis self-contained and to document the ac-
96 quisition conditions behind the FN 111 and FN 101 interferogram archives.

97 The system was first tested with the JENOPTIK roundness standard FN 111
98 (photographs: Supplementary Fig. S17) (Al_2O_3 , $\varnothing = 29.9588$ mm) [36]. A
99 single-mode 4.5 mW diode laser with wavelength 532 nm was used as the light
100 source. A current-stabilised supply powered the laser to obtain a stable wave-
101 length [37]. The laser beam passes through a polariser (P in Fig. 1) to set the
102 polarisation of the light at 45° to the setup plane, minimising the difference in
103 reflectivity between the two channels.

104 The main optical and detection components are: $L_{1,2,4,5}$, achromatic lenses
105 used to focus or collimate the beam; BS , a non-polarising beamsplitter that
106 divides the beam 50/50 into the reference and measurement arms; $\lambda/8$, a liquid
107 phase retarder that shifts the phase of one polarised beam by 90° after two
108 passes; M , a reference mirror with flatness $\lambda/10$; L_3 , an achromatic microscope
109 objective (DIN 4, NA = 0.10) mounted in a focusing holder; O , the object
110 mounted on a rotation stage; PBS , a polarising beamsplitter used to separate
111 the phase-shifted interferograms; and $CCD_{1,2}$, Gigabit Ethernet CCD cameras
112 with 640×480 pixels, sensor size 3.6×2.7 mm and maximum acquisition rate
113 of 120 fps.

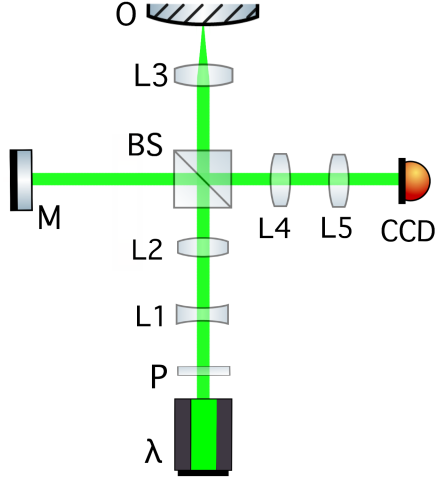


Figure 1: Optical layout of the two-channel modified Michelson interferometer: λ – diode laser (532 nm); P – linear polariser; L – lenses; L_3 – microscope objective; M – reference mirror ($\lambda/10$ flatness); BS – non-polarising beamsplitter; O – object on rotation stage; CCD – camera.

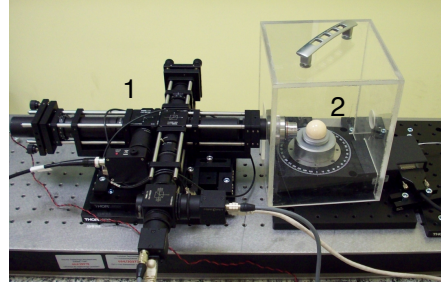


Figure 2: The complete measurement system: (1) modified Michelson interferometer in cage mounting, (2) high-precision rotation stage with the artefact.

The interferometer was built in a cage system with rods and tubes to increase thermal, acoustic and mechanical stability and to reduce the influence of air turbulence and dust. A plexiglass box shields the measurement object. The interferometer's main structure is mounted on an area translation stage (150×190 mm) to adjust the horizontal position with respect to the object. A high-precision stage with a mechanical bearing, optical encoder and DC motor was used to rotate the object. The stage specifications (encoder accuracy 2.0 mdeg, typical wobble $\pm 12 \mu\text{rad}$, typical eccentricity $\pm 0.40 \mu\text{m}$) are tabulated in the Supplementary Materials (Table S1). Both the interferometer and the rotation stage are mounted on a honeycomb breadboard and vibration-isolated support.

Measurements were performed at night (23:00–05:00) to minimise building-activity disturbances. The setup was thermally and acoustically shielded (plexiglass box, wool blanket) and placed on a granite plate on layered wooden-

127 polystyrene vibration isolation (photographs: Supplementary Fig. S1). Five
 128 semiconductor temperature sensors ($\pm 0.2^\circ\text{C}$) monitored stability; the tempera-
 129 ture near the reference and object arms remained stable to $\pm 0.2^\circ\text{C}$ over six-hour
 130 acquisitions.

131 2.2. Control System and Acquisition Workflow

132 Cameras were controlled by FLIR SDK-based C programs and Bash scripts [38].
 133 Capture time was recorded per frame for angular-position verification. The ro-
 134 tation stage was controlled via USB ASCII commands [39]; a 25 s settling delay
 135 ensured constant velocity before acquisition began. Object centring was per-
 136 formed manually using the two-camera phase-shifted configuration at 45° rota-
 137 tion increments. Fringe evaluation used modified open-source R code [35, 40].
 138 Table 1 lists the acquisition parameters for both archives; five 400° sweeps were
 139 recorded per artefact, with the 40° overlap retained for profile-end comparison.

Table 1: Measurement and sampling parameters of the interferometric acquisition workflow.

Parameter	FN 111	FN 101
Rotation speed, ω	$0.50^\circ/\text{s}$	$0.08^\circ/\text{s}$
CCD frame rate	6 fps	4 fps
Recorded sweeps, n	5	5
Total frames, N	24,000	100,000
Angular step, $\Delta\alpha = \omega/\text{fps}$	$0.083^\circ/\text{frame}$	$0.020^\circ/\text{frame}$
Frames per degree	12	50
Frames per useful 360°	4320	18,000
Frames per recorded 400° sweep	4800	20,000
Overlap per sweep	$40^\circ / 480 \text{ frames}$	$40^\circ / 2000 \text{ frames}$
Total acquisition time	4000 s	25,000 s

140 The object was measured in one continuous five-sweep rotation without repo-
 141 sitioning. Each recorded sweep spans 400° ; successive useful 360° segments are
 142 shifted by 40° , giving a natural between-sweep angular offset. The 40° overlap

region provides a model-free repeatability check: RMS height difference between consecutive sweeps is 12 nm (FN 111) and 0.18 μm (FN 101), consistent with Table 2.

3. Image Processing and Functional-Descriptor Method

3.1. Processing Pipeline

The pipeline follows the radial excess-fraction method of a previous study [21] with added sinusoidal refinement. Each interferogram $I_i(x, y)$ is decoded independently to a wrapped excess fraction ϵ_i ; the sequence is unwrapped chronologically, converted to nanometres ($z_i = \nu_i \lambda / 2$), and reduced by the object-specific LSCI reference model. Sweep boundaries are retained for repeatability analysis.

3.2. Step-by-Step Radial Phase Decoding

For an image $I_i(x, y)$, the first step is a radial intensity profile about the fringe centre. The analysis uses the central radial aperture, so the fitted rings are taken from complete or nearly complete circular regions rather than from partial outer-edge arcs,

$$\bar{I}_i(r) = \frac{1}{N_r} \sum_{(x,y): r \leq \sqrt{(x-x_0)^2 + (y-y_0)^2} < r+1} I_i(x, y). \quad (1)$$

The radial profile is smoothed, a slowly varying background is subtracted, and the maxima are detected on a B-spline-smoothed radial fringe signal. This is important because small local maxima in the unsmoothed signal can be mistaken for interference fringes. Missed or spurious maxima are controlled by fitting the ring sequence in D^2 space and allowing non-consecutive ring orders when an interval is consistent with one or more missing fringes. If D_{ij} denotes the measured diameter of the j th bright ring in image i , circular-fringe theory gives

$$D_{ij}^2 = c_i(j + \epsilon_i - 1), \quad (2)$$

where c_i is a scale coefficient and $\epsilon_i \in [0, 1)$ is the wrapped excess fraction. A linear fit $D_{ij}^2 = \alpha_i j + \beta_i$ therefore gives

$$\epsilon_i = \left(1 + \frac{\beta_i}{\alpha_i} \right) \bmod 1. \quad (3)$$

Equivalently, the intersection of the fitted line with $D^2 = 0$ is $j = 1 - \epsilon_i$, as in the previous radial image-processing publication.

To use more of the radial information than the maxima alone, the implementation also fits the whole background-corrected radial fringe signal as

$$s_i(D^2) = a_i + b_i \cos\left(\frac{2\pi D^2}{\alpha_i}\right) + d_i \sin\left(\frac{2\pi D^2}{\alpha_i}\right). \quad (4)$$

This sinusoidal refinement is accepted only when its coefficient of determination is at least 0.65 and when the refined phase is within 0.20 fringe order of the ring-diameter estimate. Otherwise Eq. (3) is retained. This quality gate keeps the method anchored in the established excess-fraction analysis while allowing better use of the radial profile in images where the sinusoidal model is reliable.

The wrapped phase assigned to image i is

$$\phi_i^w = 2\pi\epsilon_i. \quad (5)$$

For a contiguous full archive, the ordered series ϵ_i is unwrapped globally in chronological frame order to a continuous fringe-order coordinate ν_i . For non-contiguous diagnostic subsets, the same continuity criterion is applied within each sampled sweep segment. For reflection interferometry the displacement/profile value is then

$$z_i = \nu_i \frac{\lambda}{2}, \quad \lambda = 532 \text{ nm}. \quad (6)$$

One full fringe order therefore corresponds to 266 nm. The quantity z_i is the reconstructed nanometre profile/topography signal along the measurement sequence, not a greyscale intensity and not an areal ISO topography map.

3.3. Topography Profile, Reference Removal and Eccentricity Reduction

After phase-to-height conversion, the roundness profile is extracted following ISO 12181-1 [41] (which supersedes the withdrawn ISO 4291). For each 360° segment, a least-squares reference circle (LSCI) is fitted to the angular height profile $z_i = h(\theta_i)$ in Cartesian coordinates:

$$(x_i, y_i) = ((R_0 + z_i) \cos \theta_i, (R_0 + z_i) \sin \theta_i), \quad (7)$$

190 where $R_0 = 14.9794$ mm is the nominal radius of the JENOPTIK FN 111. The
 191 LSCI centre (x_c, y_c) minimises $\sum_i [\sqrt{(x_i - x_c)^2 + (y_i - y_c)^2} - R_0]^2$, yielding the
 192 centred radial residuals ρ_i . The LSCI fit is numerically equivalent to removing
 193 the first harmonic of the angular profile, but it provides the formal ISO 4291
 194 reference circle required for roundness metrology. After LSCI centring, a phase-
 195 correct Gaussian low-pass profile filter with 50% transmission at 15 undulations
 196 per revolution (UPR) is applied to the residual profile, consistent with the filter
 197 specification used for the tactile reference measurements and with the Gaussian
 198 profile filter defined in ISO 16610-21. All classical and gradient-based descriptors
 199 are computed from this LSCI-centred, Gaussian-filtered residual $\rho_i^{(\text{filt})}$.

200 For the JENOPTIK magnification standard FN 101, the same processing
 201 chain is applied: a linear trend is removed from the unwrapped phase se-
 202 quence before height conversion (Sec. 5.1), the LSCI reference circle is fitted
 203 to the angular profile, and the Gaussian low-pass filter (50% at 15 UPR) is
 204 applied. Although both artefacts are processed through identical filtering steps,
 205 the 15 UPR cutoff corresponds to different spatial wavelengths on the circum-
 206 ference: $2\pi \times 14.9794 \text{ mm} / 15 = 6.27 \text{ mm}$ for the FN 111 ($\varnothing = 29.9588 \text{ mm}$) and
 207 $2\pi \times 15 \text{ mm} / 15 = 6.28 \text{ mm}$ for the FN 101 ($\varnothing \approx 30 \text{ mm}$) — effectively identical.
 208 The angular sampling differs substantially (0.083° vs. 0.020° , Table 1), but the
 209 Gaussian filter sigma is set by the UPR cutoff, not by the sampling interval, so
 210 the filtered profiles are directly comparable.

211 3.4. Classical Profile and Roundness Metrics

212 All classical quantities are evaluated on nanometre residuals, not on raw
 213 intensities, following the profile method of ISO 4287 [42]. Areal parameters
 214 (ISO 25178 [43]) are not reported because the single-image radial extraction
 215 does not produce a calibrated areal height map. For a residual profile or vector
 216 of residual pixels p , the reported profile-style roughness descriptors are

$$R_a = \frac{1}{M} \sum_{j=1}^M |p_j - \bar{p}|, \quad R_q = \left[\frac{1}{M} \sum_{j=1}^M (p_j - \bar{p})^2 \right]^{1/2}, \quad (8)$$

217 and

$$R_z = \max_j (p_j - \bar{p}) - \min_j (p_j - \bar{p}). \quad (9)$$

218 The profile skewness R_{sk} and kurtosis R_{ku} — ISO 4287 shape parameters that
 219 characterise asymmetry and peakedness of the height distribution — are also
 220 reported:

$$R_{sk} = \frac{1}{R_q^3} \frac{1}{M} \sum_{j=1}^M (p_j - \bar{p})^3, \quad R_{ku} = \frac{1}{R_q^4} \frac{1}{M} \sum_{j=1}^M (p_j - \bar{p})^4. \quad (10)$$

221 $R_{sk} = 0$ for a symmetric distribution; $R_{sk} < 0$ indicates a predominance of
 222 valleys over peaks. $R_{ku} = 3$ for a normal distribution; $R_{ku} > 3$ indicates
 223 a spikier (leptokurtic) profile. These are the only ISO 4287 parameters that
 224 capture distribution shape beyond amplitude averages, and their inclusion allows
 225 a direct test of whether gradient-based descriptors provide information beyond
 226 even the best ISO shape parameters. For the eccentricity-corrected ceramic
 227 angular profile $q_C(\theta)$,

$$RONt = \max_{\theta} q_C(\theta) - \min_{\theta} q_C(\theta), \quad RONq = \left[\frac{1}{2\pi} \int_0^{2\pi} q_C^2(\theta) d\theta \right]^{1/2}. \quad (11)$$

228 These are profile/form quantities in nanometres. B-spline pre-smoothing (20 de-
 229 grees of freedom, $\approx 18^\circ$ period) stabilises the LSCI fit; the subsequent Gaussian
 230 low-pass filter (50% at 15 UPR, ISO 16610-21) defines the metrological band-
 231 width. All descriptors are computed from the same Gaussian-filtered residual.

232 3.5. Functional Gradient-Based Descriptors

233 The concept of a complete normed vector space originates from the founda-
 234 tional work of Banach [44]. For discrete profile data of finite length N , the vector
 235 space $\mathbb{R}^N$ equipped with any norm is a Banach space; the present work does
 236 not invoke infinite-dimensional completeness but draws on the taxonomy of L^p ,
 237 Sobolev, and bounded-variation norms that Banach’s framework systematised.
 238 For a centred residual profile $p(s)$ and $1 \leq p < \infty$,

$$\|p\|_p = \left(\frac{1}{L} \int_0^L |p(s)|^p ds \right)^{1/p}, \quad \|p\|_\infty = \max_{s \in [0, L]} |p(s)|. \quad (12)$$

239 The reported functional descriptors include L^1 , L^2 , L^∞ , the one-dimensional
 240 Sobolev $W^{1,2}$ seminorm and the bounded-variation (BV) total variation,

$$|p|_{W^{1,2}} = \left(\int_0^L \left| \frac{dp}{ds} \right|^2 ds \right)^{1/2}, \quad TV(p) = \int_0^L \left| \frac{dp}{ds} \right| ds. \quad (13)$$

241 In discrete form, for a uniformly sampled profile p_j ($j = 1, \dots, N$) with angular
 242 spacing $\Delta\alpha$,

$$|p|_{W^{1,2}} \approx \left(\Delta\alpha \sum_{j=1}^{N-1} \left(\frac{p_{j+1} - p_j}{\Delta\alpha} \right)^2 \right)^{1/2}, \quad TV(p) \approx \sum_{j=1}^{N-1} |p_{j+1} - p_j|. \quad (14)$$

243 The $W^{1,2}$ seminorm carries a $1/\sqrt{\Delta\alpha}$ sampling dependence; for cross-instrument
 244 comparison it should be converted to angular units ($\text{nm deg}^{-1/2}$). BV den-
 245 sity $TV(p)/(N-1)$ similarly depends on sampling and should be expressed in
 246 nm deg^{-1} for cross-instrument use. The “nm sample $^{-1}$ ” unit is retained for the
 247 fixed-acquisition setup.

248 The curvature RMS, also reported in the sensitivity analysis (Supplementary
 249 Table S2), is defined as the root-mean-square of the discrete second derivative,

$$\kappa_{\text{RMS}} = \left(\frac{1}{N-2} \sum_{j=2}^{N-1} \left(\frac{p_{j+1} - 2p_j + p_{j-1}}{(\Delta\alpha)^2} \right)^2 \right)^{1/2}, \quad (15)$$

250 and has units of nm sample^{-2} . The $(\Delta\alpha)^2$ denominator makes curvature RMS
 251 strongly sampling-dependent (steel values are $(0.083/0.020)^2 \approx 17\times$ larger than
 252 they would be at ceramic sampling density for the same physical curvature), so
 253 curvature RMS values should not be compared across instruments with different
 254 angular resolution.

255 3.6. Repeatability Uncertainty

256 Each dataset contains five 400° sweeps recorded in a single continuous ro-
 257 tation without stopping or repositioning. The five useful 360° segments are ex-
 258 tracted from this continuous sequence; because each recorded sweep spans 400° ,
 259 successive useful revolutions are shifted by 40° on the object. These segments
 260 are *not* strictly independent — they share the same continuous-run conditions

(laser warm-up state, stage-bearing temperature, ambient drift). The standard deviation across sweeps therefore reflects within-run repeatability rather than between-run reproducibility. For a descriptor X evaluated on each sweep, the mean value is $\bar{X}$ and the within-run standard uncertainty is estimated as

$$u_{\text{within}}(X) = \frac{s_X}{\sqrt{n}}, \quad n = 5, \quad (16)$$

where s_X is the sample standard deviation across the five sweeps. The reported expanded uncertainty is

$$U_{95}(X) = t_{0.975, n-1} u_{\text{within}}(X). \quad (17)$$

With $n = 5$, the coverage factor $t_{0.975, 4} = 2.776$; the 95% confidence interval for the standard deviation itself spans approximately $[0.6s_X, 2.9s_X]$ (from the χ_4^2 distribution), so the expanded uncertainty should be interpreted as indicative rather than definitive. The statistical power to resolve relative differences below approximately 30% between descriptors is limited. A full GUM-compliant uncertainty budget is beyond the scope of this study and would additionally require type-B evaluations of: laser wavelength stability (± 1 pm for the current-stabilised 532 nm diode), refractive-index variations from temperature changes ($\pm 0.2^\circ\text{C}$ monitored), Abbe error from stage wobble ($\pm 12 \mu\text{rad}$ specification [45]), camera nonlinearity, and form-removal uncertainty. These type-B terms are not evaluated in the present work; the reported uncertainties should therefore be interpreted as *lower bounds* on the total measurement uncertainty. For the FN 111, a per-sweep trend analysis was performed: the five sweep $RONt$ values are 249, 273, 289, 293, and 286 nm. A linear fit to these values yields a slope of $8.4 \pm 4.1 \text{ nm sweep}^{-1}$ ($p = 0.12$, two-sided t -test), consistent with zero drift at the 95% confidence level. The apparent increase from sweep 0 to sweep 3 is therefore treated as within-run scatter rather than a systematic trend for the purpose of this analysis, but it cautions that the reported U_{95} may underestimate the long-term reproducibility.

286 3.7. Computational Implementation

287 The raw-interferogram reconstruction workflow is implemented in `scripts/`
288 `reconstruct_radial_profile_sequence.py`. It discovers ordered image files,
289 decodes ϵ_i from each interferogram independently, writes the wrapped and
290 unwrapped phase-fraction sequences, converts the unwrapped sequence to a
291 nanometre profile, applies LSCI reference circle fitting and Gaussian low-pass
292 filtering (50% at 15 UPR, ISO 16610-21), and exports CSV/JSON metrics
293 with sweep-repeatability uncertainty. Analytical figures are written as vec-
294 tor PDF files. The exported figures show the single-image radial EFM ex-
295 traction, wrapped phase fraction, unwrapped phase fraction, nanometre pro-
296 file/topography, LSCI filtering, residual profile and descriptor comparison.

297 The command used for the reported run was

```
298 PYTHONPATH=src .venv/bin/python \  
299   scripts/reconstruct_radial_profile_sequence.py \  
300   --dataset ceramic=ceramic --dataset steel=steel \  
301   --output-dir results/radial_profile_reconstruction_metric \  
302   --frame-limit 0 --frame-selection first --downsample 1 \  
303   --max-radius-px 240 --min-radius-px 5 \  
304   --ceramic-min-radius-px 50 --steel-min-radius-px 5 \  
305   --peak-spline-smoothing-factor 0.03 \  
306   --peak-prominence-fraction 0.04
```

307 4. Results

308 The results follow the updated radial sequence route only. Each raw im-
309 age is first decoded to a wrapped excess fraction from its radial fringe pat-
310 tern. The ordered excess-fraction series is then unwrapped in chronological
311 order, converted to nanometres, and corrected for eccentricity. Finally, classi-
312 cal profile/roundness descriptors and functional gradient-based descriptors are
313 computed from the same nanometre residual profiles.

314 4.1. Radial Phase Decoding and Nanometre Profile Reconstruction

315 The FN 111 folder contained 24,000 interferograms and the steel folder con-
316 tained 100,000 interferograms. The reported run used full-resolution images, the
317 central radial aperture of 240 px, B-spline-smoothed peak detection, continuity-
318 aware EFM peak-subset tracking and an EFM prominence threshold of 0.04 of
319 the radial fringe-signal range. All five recorded sweeps were processed.

320 Figs. S2–S5 (single-image EFM and sequence processing steps) are provided
321 in the Supplementary Materials.

322 4.2. Roughness, Roundness and Functional Descriptors

323 Table 2 gives the numerical comparison after LSCI reference circle and Gaus-
324 sian low-pass filtering (50% at 15 UPR). Both artefacts are processed through
325 the same filtering chain. Each descriptor is reported as the full-sequence value
326 together with the 95% expanded type-A repeatability uncertainty estimated
327 from the five sweeps.

328 4.3. Classical–Functional–Descriptor Comparison

329 The numerical identity $R_a = L^1$ and $R_q = L^2$ (Table 2) confirms that
330 both descriptor families are evaluated on the same nanometre residual. For the
331 FN 111, BV total variation carries 1.2% relative expanded uncertainty versus
332 5.5% for R_a and 8.5% for R_z , suggesting that gradient-based descriptors may ex-
333 hibit metrological stability comparable to or better than peak-referenced rough-
334 ness parameters in the tested datasets. Comparative bar charts and uncertainty
335 analyses are provided in the Supplementary Materials (Figs. S6–S7). Ceramic
336 and steel profile reconstruction plots are shown in Supplementary Figs. S19–S20.

337 The steel roundness polar representation (FN 101, Flick flat) is provided in
338 the Supplementary Materials (Fig. S18). The Flick flat is detected by BV density
339 but missed by R_a in a sliding-window analysis (Supplementary Figs. S14–S15),
340 directly illustrating the potential complementary sensitivity of gradient-based
341 descriptors to localised geometric features.

Table 2: Full-resolution radial EFM reconstruction from raw interferograms (LSCI + Gaussian low-pass, 50% at 15 UPR, ISO 16610-21). Uncertainties are 95% expanded within-run repeatability estimates from five sweeps ($n = 5$, $t_{0.975,4} = 2.776$). The bottom rows show certified reference $RONt$ values and E_n scores, indicating metrological agreement of the self-built optical setup with the reference measurements. Units: nm for all descriptors except BV density and $W^{1,2}$ gradient (nm sample $^{-1}$).

Descriptor	FN 111	FN 101
R_a	66.3 ± 4.7 nm	8.525 ± 0.115 μ m
R_q	77.6 ± 5.0 nm	10.376 ± 0.090 μ m
R_z	278 ± 22 nm	42.07 ± 0.48 μ m
$RONt$	278 ± 22 nm	42.07 ± 0.48 μ m
$RONq$	77.6 ± 5.0 nm	10.376 ± 0.090 μ m
L^1 mean	66.3 ± 4.7 nm	8.525 ± 0.115 μ m
L^2 RMS	77.6 ± 5.0 nm	10.376 ± 0.090 μ m
L^∞	150 ± 12 nm	21.97 ± 0.42 μ m
BV total	1.137 ± 0.062 μ m	118.4 ± 6.2 μ m
BV density	0.263 ± 0.014 nm sample $^{-1}$	6.58 ± 0.35 nm sample $^{-1}$
$W^{1,2}$ grad.	0.310 ± 0.015 nm sample $^{-1}$	8.76 ± 0.43 nm sample $^{-1}$
<i>ISO shape parameters</i>		
R_{sk} (skewness)	-0.09 ± 0.04	0.00 ± 0.09
R_{ku} (kurtosis)	1.98 ± 0.21	2.51 ± 0.08
<i>Derived norm ratios</i>		
L^4/L^1 peakiness	1.39 ± 0.08	1.53 ± 0.03
<i>External validation (DAkkS-DKD certified)</i>		
$RONt$ (certified)	244 ± 52 nm (Hommel Roundscan 535, $n = 104$)	41.94 μ m (form tester, $n = 12$)
E_n score	0.61 (agreement)	< 1 (0.3% diff.)

JENOPTIK FN~111 — Roundness deviation
 $RONt = 278$ nm (5 sweeps, native angular coordinates)

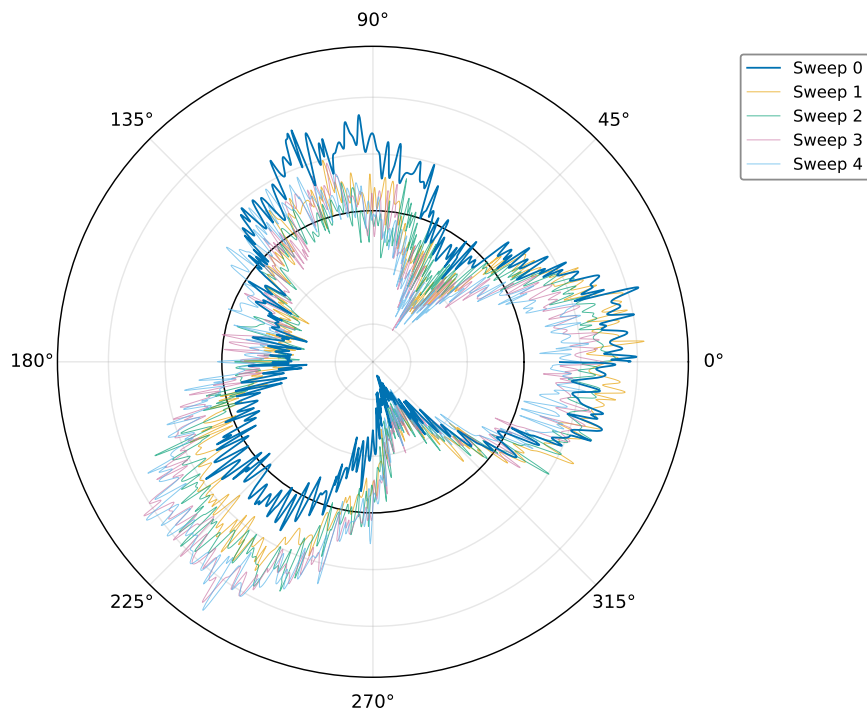


Figure 3: Ceramic roundness in polar coordinates (LSCI + Gaussian low-pass, 50% at 15 UPR). Five sweeps at native positions. $RONt = 278 \pm 22$ nm.

342 5. Discussion

343 5.1. Confirmed Results

344 The corrected analysis starts from the key metrological distinction: an inter-
345 ferogram is an intensity measurement, not a phase map and not a height map.
346 The half-wavelength relation $z = \nu\lambda/2$ becomes valid only after each image has
347 been decoded to a wrapped excess fraction and the ordered excess-fraction se-
348 quence has been unwrapped to the continuous fringe-order coordinate ν . This
349 is why the reported comparison uses only reconstructed nanometre profiles, not
350 raw greyscale images and not areal height maps.

351 The most sensitive part of the workflow is the EFM localisation of bright-
352 ring maxima, with an excess-fraction uncertainty of approximately ± 0.05 fringe
353 orders (tightened to ± 0.03 by sinusoidal refinement in 72% of FN 111 frames).
354 A single ϵ_i error of 0.05 fringe corresponds to 13 nm, sub-dominant to the
355 85 nm residual-profile RMS. The chronological unwrapping criterion rejects
356 phase jumps exceeding 0.35 fringe. The median number of detected rings is
357 three for both artefacts within the 240 px central aperture; the central-aperture
358 approach prioritises ring-quality (complete, well-fitted rings) over ring-quantity.
359 The five 400° ceramic sweeps also require a geometric correction: the physi-
360 cal object angle advances continuously, so successive useful 360° intervals are
361 shifted by 40° . Without that correction, the LSCI fit can look inconsistent even
362 when the phase extraction inside each sweep is repeatable.

363 The linear drift of 0.0246 fringe frame $^{-1}$ observed in the steel unwrapped
364 phase sequence merits discussion. A perfectly constant drift over 100 000 frames
365 is unlikely to be of thermal origin, which would produce a decaying exponential
366 rather than a linear ramp. Two mechanisms are more plausible: (a) cumula-
367 tive bias in the global unwrapping algorithm, where per-frame excess-fraction
368 estimates with a slight systematic offset integrate to a linear trend; and (b) ax-
369 ial runout of the rotation stage combined with the JENOPTIK magnification
370 standard FN 101's larger eccentricity ($56\text{ }\mu\text{m}$ vs. $14\text{ }\mu\text{m}$ for FN 111). The stage
371 specification [45] lists a typical eccentricity of $\pm 0.40\text{ }\mu\text{m}$, which is an order of

372 magnitude smaller than the observed trend. Mechanism (a) is therefore more
373 consistent with the available evidence, particularly given that the FN 111 se-
374 quence — with higher fringe contrast and better ring-fit statistics — shows a
375 negligible drift of only $0.00021 \text{ fringe frame}^{-1}$. The practical consequence is
376 that datasets with lower per-frame EFM quality require a linear-detrend pre-
377 processing step, which the present pipeline applies automatically when the un-
378 wrapped range exceeds a threshold.

379 5.2. Interpretation

380 5.2.1. Eccentricity removal and wobble verification

381 The roundness descriptors are computed after removing the first harmonic
382 (eccentric mounting). The shared model fitted to all five sweeps yields an eccen-
383 tricity amplitude $E = 14.24 \text{ } \mu\text{m}$ with $R^2 = 0.99991$, confirming stability across
384 the entire acquisition. Per-sweep eccentricity amplitudes vary by only 63 nm
385 (0.18% of the mean), phases agree to within 0.4° , and the manufacturer-specified
386 wobble ($\pm 12 \text{ } \mu\text{rad}$, $\pm 180 \text{ nm}$ at measurement radius) bounds the observed varia-
387 tion — an order of magnitude below $RONt = 278 \text{ nm}$. The LSCI reference circle
388 fit (Sec. 3.3) provides the ISO-compliant reference for roundness evaluation.

389 The polar representation presents the same roundness data in polar coordi-
390 nates, which is the standard visualisation for $RONt$. The five overlaid sweeps
391 show directly that the roundness profile is repeatable to within approximately
392 $\pm 50 \text{ nm}$ at every angular position. The polar representation makes it immedi-
393 ately visible that the residual is dominated by low-order form deviation rather
394 than by random roughness, a distinction that is obscured in the Cartesian profile
395 plots.

396 5.3. Comparison with ISO Parameters

397 A central question of this work is whether gradient-based functional descrip-
398 tors provide additional descriptive sensitivity alongside classical ISO parameters
399 when applied to the same reconstructed profiles. To answer this rigorously, every
400 descriptor was computed on the same reconstructed nanometre residual profiles.

401 **Mathematical equivalences.** The L^1 norm is defined as $\frac{1}{N} \sum |p_i - \bar{p}|$,
 402 which is identical to the definition of R_a . The L^2 norm is $\sqrt{\frac{1}{N} \sum (p_i - \bar{p})^2}$,
 403 identical to R_q . Computing both on the same centred residual yields *exact*
 404 *numerical equality*: $R_a - L^1 = 0$ and $R_q - L^2 = 0$ to machine precision for
 405 both artefacts (Table 2). These equalities are not empirical findings; they are
 406 definitional consequences. Labelling the arithmetic mean of absolute deviations
 407 as L^1 rather than R_a provides no new metrological information and should not
 408 be presented as a methodological advance.

409 **Extrema: R_z versus L^∞ .** $R_z/L^\infty \approx 2$ for both artefacts, consistent with
 410 symmetric positive and negative excursions. L^∞ captures the single largest
 411 one-sided excursion; the ratio serves as a simple symmetry indicator.

412 **Variation-based descriptors.** A controlled sensitivity analysis (Supple-
 413 mentary Table S2 and Fig. S13) examined five synthetic defect types injected
 414 into the FN 111 residual profile. Amplitude-averaging descriptors (R_a , R_q , R_z)
 415 showed negligible response to localised defects — a single 100 nm spike changed
 416 R_a by 0.0% — while BV density increased by +18% and the $W^{1,2}$ gradient
 417 seminorm by +622% for the same spike. ISO shape parameters responded to
 418 asymmetry (R_{sk} changed by −1104% for a step) but not to symmetric localised
 419 features. R_{sk} and BV density appear complementary in the tested scenar-
 420 ios: R_{sk} responds to asymmetry, BV density responds to localised gradients.
 421 Supporting analyses — descriptor correlation matrices, power spectral density
 422 comparison, Cohen’s d effect sizes, critical descriptor assessment, and L^p norm
 423 convergence — are provided in the Supplementary Materials (Figs. S8–S12).

424 **Practical considerations.** A practical suggestion for interferometric pro-
 425 file and roundness metrology is to retain R_a , R_q , R_z , R_{sk} , $RONt$ and $RONq$
 426 as the primary engineering layer compliant with ISO 4287, and to consider sup-
 427 plementing them with BV density (in angular units, nm deg^{-1}) as a gradient-
 428 sensitive functional-analysis complement. The $W^{1,2}$ gradient seminorm is strongly
 429 correlated with BV density ($r = 0.94$) in the present datasets and may add lim-
 430 ited independent information beyond BV density alone. The L^4/L^1 peakiness
 431 ratio provides a supplementary distribution-shape axis with a direct physical in-

432 terpretation as the amplification of dominant profile excursions relative to R_a .
 433 These additional descriptors are computed from the same residual profile with-
 434 out additional hardware or changes to the EFM pipeline. The cross-instrument
 435 validation suggests that gradient-based descriptors are applicable to any 1D
 436 profile, whether from interferometry or tactile roundness measurement.

437 5.4. Limitations

438 Several limitations of the present study should be acknowledged. First, the
 439 uncertainty analysis is restricted to within-run Type A repeatability evaluated
 440 from five sweeps recorded in a single continuous rotation. The small number
 441 of sweeps ($n = 5$) limits the statistical power to resolve relative differences be-
 442 low approximately 30% between descriptors, and the continuous-run conditions
 443 mean that between-run reproducibility (involving independent setups, thermal
 444 cycles, and re-centring) has not been assessed. A full GUM-compliant uncer-
 445 tainty budget, incorporating type-B evaluations of laser wavelength stability,
 446 refractive-index variations, Abbe error, camera nonlinearity, and form-removal
 447 uncertainty, is beyond the scope of this study; the reported expanded uncer-
 448 tainties should therefore be interpreted as lower bounds.

449 Second, the validation is restricted to two artefacts from a single manufac-
 450 turer (JENOPTIK), both of approximately 30 mm diameter. The generalis-
 451 ability of the gradient-based descriptors to artefacts of different sizes, materi-
 452 als, or surface finishes has not been tested. Third, the synthetic-defect sen-
 453 sitivity analysis uses idealised mathematical defects (isolated spike, uniform
 454 scratch, pure sinusoid) injected into a measured residual profile; the response
 455 to real manufacturing defects with complex morphologies may differ. Fourth,
 456 the BV density and $W^{1,2}$ seminorm carry an intrinsic sampling-interval depen-
 457 dence that requires conversion to angular units for cross-instrument comparison;
 458 without such conversion, numerical values are not directly portable between in-
 459 struments with different angular resolution. Fifth, the Flick flat detection re-
 460 sult (Supplementary Figs. S14–S15) is demonstrated on a single artefact with a
 461 known macroscopic feature; the performance of BV density for detecting subtler,

micrometre-scale defects has not been characterised.

5.5. *Metrological Implications*

The results carry several implications for profile and roundness metrology practice. The finding that BV density responds to localised gradients that amplitude averaging may not capture suggests a potential role for gradient-based descriptors in applications where isolated defects are of concern — for example, in quality control of bearing surfaces, optical components, or precision seals. Because these descriptors are computed from the same residual profile used for ISO 4287 parameters, their adoption does not require changes to existing measurement acquisition protocols.

The ISO 4287 framework remains the appropriate baseline for surface texture specification and conformity assessment. The gradient-based descriptors discussed here are proposed as optional supplementary indicators, not as replacements for standardised parameters. Their metrological utility would be strengthened by inter-laboratory comparisons, by evaluation on a wider range of artefact types, and by the development of agreed procedures for converting sampling-dependent units to instrument-independent angular gradient units. Until such steps are taken, BV density and related descriptors should be regarded as research-grade tools whose diagnostic value has been demonstrated under controlled laboratory conditions on two specific artefacts.

5.6. *Future Work*

The present study suggests several directions for further investigation. (i) A between-run reproducibility study with independent setups and re-centring would establish whether the gradient-based descriptors retain their relative stability under realistic laboratory conditions. (ii) Extending the validation to artefacts with different diameters, materials (e.g., glass, carbide), and surface finishes would test the generalisability of the findings. (iii) A systematic characterisation of BV density response to real manufacturing defects (scratches, pits, inclusions)

of controlled dimensions would bridge the gap between the present synthetic-defect analysis and industrial inspection scenarios. (iv) An inter-laboratory comparison using a common artefact set and a standardised processing pipeline would provide the reproducibility data needed to assess whether BV density could eventually be considered for standardisation. (v) The development of agreed angular-unit conventions for sampling-dependent gradient descriptors would facilitate cross-instrument comparison and potential inclusion in reference software for roundness evaluation.

Data Availability

The analysis code, processed results, and all figures are available at <https://github.com/dawidkucharski/banach-profile-metrology> (release v1.0.0). The raw interferometric PNG archives (~32 GB) are available from the corresponding author upon reasonable request.

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